Logistic Regression

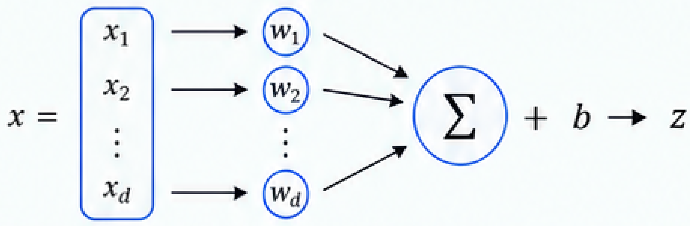
Clean notes from handwritten lecture pages

1 Overview

Logistic regression is a linear classifier used for binary classification. It models the probability that an input belongs to class 1.

2 Linear score

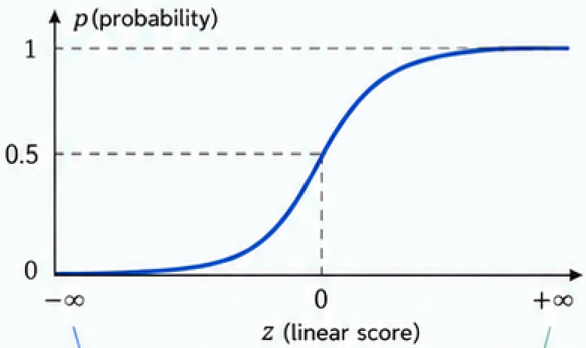
$$z = w^T x + b$$



z is a linear score (can be any real number).

3 Sigmoid function

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$



As $z \to -\infty$, $p \to 0$

As $z \to +\infty$, $p \to 1$

Output range: $0 < p < 1$

4 Prediction rule

Use a threshold of 0.5:

If $p \geq 0.5$, predict class 1

Otherwise, predict class 0

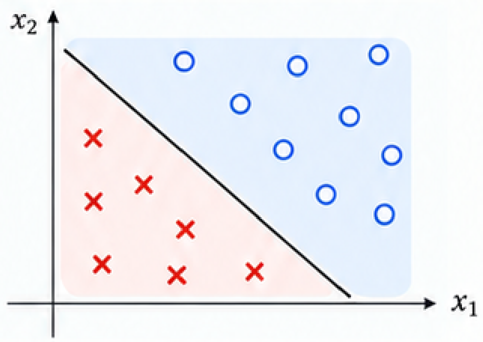
5 Why not plain linear regression?

Aspect	Linear Regression	Logistic Regression
Output range	$(-\infty, +\infty)$ Can be any real number	$(0, 1)$ Always a valid probability
Suitability for classification	✗ Not suitable	✓ Well-suited

6 Sample training data

x_1	x_2	y (label)
2	1	1
1	2	1
−1	−1	0
−2	−1	0
1	−1	1

7 Linear decision boundary



○ Class 1 $(y = 1)$

✗ Class 0 $(y = 0)$

— Decision boundary $(z = w^T x + b = 0)$

Logistic regression is a linear classifier.

Key takeaway: Logistic regression converts a linear score into a probability using the sigmoid function, then classifies using a threshold.

Odds, Logit, and the Sigmoid

How logistic regression turns a linear score into a probability

1 Probability model

$$p = P(y = 1 \mid x),$$

where p is the probability of class 1.

2 Odds

$$\text{odds} = \frac{p}{1-p}$$

odds > 1 favors class 1

odds < 1 favors class 0

3 Log-odds (logit)

$$\log\left(\frac{p}{1-p}\right) = z = w^T x + b$$

The log-odds are modeled as a linear function of the input x.

4 From logit to sigmoid

$$\log\left(\frac{p}{1-p}\right) = z$$

Start from the logit definition.

$$\frac{p}{1-p} = e^z$$

Exponentiate both sides.

$$p = e^z(1-p)$$

Multiply both sides by $(1-p)$.

$$p + pe^z = e^z$$

Expand the left side.

$$p(1+e^z) = e^z$$

Factor out p.

$$p = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

Divide both sides by $(1+e^z)$ and use $e^{-z} = 1/e^z$.

$$p = \sigma(z) = \frac{1}{1+e^{-z}}$$

This is the sigmoid (logistic) function.

5 Odds examples

p	odds $= \frac{p}{1-p}$	interpretation
0.01	0.0101	about 1:99
0.50	1	equal odds
0.75	3	3:1 in favor of class 1
0.90	9	strong confidence for class 1

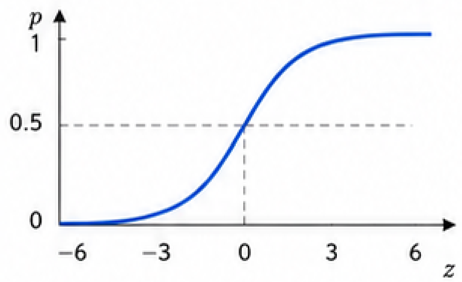
6 At the threshold

$p = 0.5$ → odds $= 1$ → log-odds $= 0$ → $z = 0$

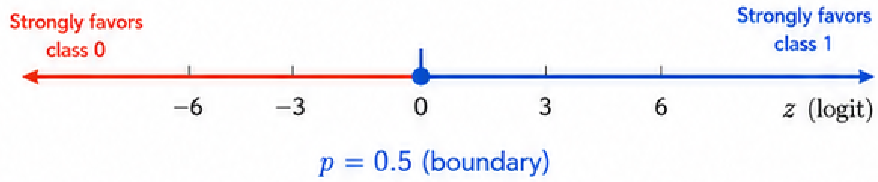
0.5 → 1 → 0 → 0

At the decision boundary, the model is maximally uncertain.

7 Logit ↔ Probability (Sigmoid)



z	−6	−3	−2	−1	0	1	2	3	6
$p = \sigma(z)$	0.002	0.047	0.119	0.269	0.500	0.731	0.881	0.953	0.998



Key takeaway: Logistic regression uses a **logit** link so a linear score can be interpreted as a probability.

Feature Example and Decision Boundary

Using features to predict a binary outcome

1 Small dataset example

Temperature-based sample data		
Min temp (x_1)	Max temp (x_2)	y (label)
20	24	0
21	25	1
24	26	1

x_1 = min temp, x_2 = max temp

2 Model

$$z = w_1 x_1 + w_2 x_2 + b$$

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

x_1 and x_2 are the features (for example, min temp and max temp).

3 Prediction rule

If $p \geq 0.5$, predict class 1

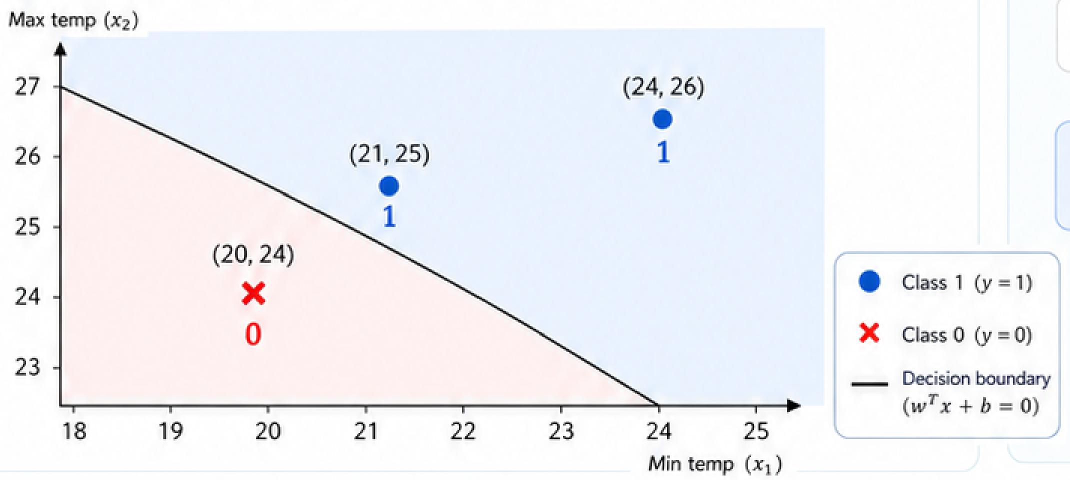
Otherwise, predict class 0

4 Decision boundary

$$w^T x + b = 0$$

This is a straight line in feature space.

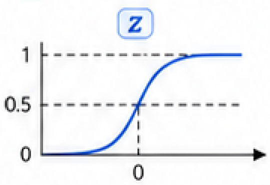
5 Decision boundary in feature space



6 Workflow

Features: $x = [x_1, x_2]$ (min temp, max temp) → **Linear score**: $z = w^T x + b$ (weighted sum + bias) → **Sigmoid**: $p = \sigma(z) = \frac{1}{1 + e^{-z}}$ (probability in (0, 1)) → **Probability**: $p \in (0,\ 1)$ (closer to 1 ⇒ more likely class 1) → **Predicted class**: if $p \geq 0.5 \Rightarrow 1$, otherwise $\Rightarrow 0$

7 Why logistic regression is a linear classifier



$p = \sigma(z)$ is nonlinear in the linear score z. → But the decision boundary $w^T x + b = 0$ is linear in the input feature space. → **Therefore, logistic regression is a linear classifier.**

Key takeaway: Logistic regression outputs probabilities in (0, 1) but separates classes with a linear decision boundary.